

Mohit Pawaskar

250840127012

Hadoop Lab Examination

Q1.



```
acts@acts-DIT400TR-55L:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as acts in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 12553. Stop it first and ensure /tmp/hadoop-acts-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 12752. Stop it first and ensure /tmp/hadoop-acts-datanode.pid file is empty before retry.
Starting secondary namenodes [acts-DIT400TR-55L]
acts-DIT400TR-55L: secondarynamenode is running as process 13019. Stop it first and ensure /tmp/hadoop-acts-secondarynamenode.pid file is empty before retry.
2025-11-10 21:50:18,462 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting resourcemanager
resourcemanager is running as process 13299. Stop it first and ensure /tmp/hadoop-acts-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 13461. Stop it first and ensure /tmp/hadoop-acts-nodemanager.pid file is empty before retry.
acts@acts-DIT400TR-55L:~$ jps
12752 DataNode
13299 ResourceManager
13461 NodeManager
33063 Jps
12553 NameNode
13019 SecondaryNameNode
acts@acts-DIT400TR-55L:~$
```

← → ↻ localhost:9870/explore.html/#/

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities ▾

Browse Directory

Show 25 entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
No data available in table								
Showing 0 to 0 of 0 entries								

150 GB Volume

Hadoop, 2025.

```
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$ hdfs dfs -mkdir -p /data_user/Mohit/
2025-11-10 21:53:47,204 WARN util.NativeCodeLoader: Unable to load native-hadoop
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$
```

```
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$ vim data.txt
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$ cat data.txt
This is Mohit Pawaskar.
My PRN is: 250840127012.
This is my TEXT file.
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$
```

```
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$ hdfs dfs -put data.txt /data_user/Mohit/
2025-11-10 21:56:11,592 WARN util.NativeCodeLoader: Unable to load native-hadoop library
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$
```

The screenshot shows the Hadoop Distributed File System (HDFS) web interface. The browser address bar indicates the URL is `localhost:9870/explore.html#/data_user/Mohit`. The main heading is "Browse Directory". Below it, the path `/data_user/Mohit` is shown. There are tabs for "Permission" and "Owner". The "Showing 1 to 1 of 1 entries" message is displayed. A modal window titled "File information - data.txt" is open, showing details for the file `data.txt`. The modal includes a "Download" button and links to "Head the file (first 32K)" and "Tail the file (last 32K)". The "Block information" section shows "Block 0" selected, with details: Block ID: 1073741825, Block Pool ID: BP-351790152-127.0.1.1-1762764837337, Generation Stamp: 1001, Size: 71, and Availability: acts-DIT400TR-55L. The "File contents" section displays the text: "This is Mohit Pawaskar. My PRN is: 250840127012. This is my TEXT file." A "Close" button is at the bottom right of the modal.

Block ID	Block Pool ID	Generation Stamp	Size	Availability
1073741825	BP-351790152-127.0.1.1-1762764837337	1001	71	acts-DIT400TR-55L

File contents:

```
This is Mohit Pawaskar.
My PRN is: 250840127012.
This is my TEXT file.
```

```
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$ hdfs dfs -cat /data_user/Mohit/data.txt
2025-11-10 21:57:50,887 WARN util.NativeCodeLoader: Unable to load native-hadoop lib
This is Mohit Pawaskar.
My PRN is: 250840127012.
This is my TEXT file.
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$
```

```
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$ hdfs dfs -ls -R /
2025-11-10 21:59:13,983 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
drwxr-xr-x - acts supergroup 0 2025-11-10 21:53 /data_user
drwxr-xr-x - acts supergroup 0 2025-11-10 21:56 /data_user/Mohit
-rw-r--r-- 1 acts supergroup 71 2025-11-10 21:56 /data_user/Mohit/data.txt
acts@acts-DIT400TR-55L:~/Mohit_Pawaskar$
```

Q2.

```
127.0.0.1 localhost
127.0.1.1 acts-DIT400TR-55L
192.168.82.191  hadoop-master
192.168.82.37   hadoop-slave-1
192.168.82.107 hadoop-slave-2
```

The following lines are desirable for IPv6 capable hosts

```
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

```
~
~
~
~
```

```
# All others are optional.  However, the defaults are probably not
# preferred.  Many sites configure these options outside of Hadoop,
# such as in /etc/profile.d
```

```
# The java implementation to use.  By default, this environment
# variable is REQUIRED on ALL platforms except OS X!
```

```
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
```

```
# The language environment in which Hadoop runs.  Use the English
# environment to ensure that logs are printed as expected.
```

```
export LANG=en_US.UTF-8
```

```
# Location of Hadoop.  By default, Hadoop will attempt to determine
# this location based upon its execution path.
# export HADOOP_HOME=
```

The following lines are desirable for IPv6 capable hosts

```
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

```
acts@acts-DIT400TR-55L:~$ mkdir -p /usr/local/hadoop_store/hdfs/datanode
```

```
acts@acts-DIT400TR-55L:~$ ls /usr/local/hadoop_store/hdfs/
```

```
datanode  namenode
```

```
acts@acts-DIT400TR-55L:~$ sudo rm -rvf /usr/local/hadoop_store/hdfs/namenode/
```

```
removed '/usr/local/hadoop_store/hdfs/namenode/in_use.lock'
```

```
removed '/usr/local/hadoop_store/hdfs/namenode/current/edits_000000000000000035-000000000000000036'
```

```
removed '/usr/local/hadoop_store/hdfs/namenode/current/edits_000000000000000077-000000000000000078'
```

```
removed '/usr/local/hadoop_store/hdfs/namenode/current/edits_0000000000000000101-0000000000000000102'
```

```
removed '/usr/local/hadoop_store/hdfs/namenode/current/edits_000000000000000069-000000000000000070'
```

```
removed '/usr/local/hadoop_store/hdfs/namenode/current/edits_000000000000000003-000000000000000004'
```

```
removed '/usr/local/hadoop_store/hdfs/namenode/current/edits_000000000000000015-000000000000000016'
```

```
removed '/usr/local/hadoop_store/hdfs/namenode/current/edits_0000000000000000115-0000000000000000116'
```



All Applications

- Cluster
- About
- Nodes
- Node Labels
- Applications
- NEW
- NEW SAVING
- SUBMITTED
- ACCEPTED
- RUNNING
- FINISHED
- FAILED
- KILLED
- Scheduler
- Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used Resources	Total Resources	Reserved Resources	Physical Mem Used %	Physical VCores Used %
0	0	0	0	0	<memory:0 B, vCores:0>	<memory:16 GB, vCores:16>	<memory:0 B, vCores:0>	7	0

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes
2	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation	Maximum Cluster Application Priority	Scheduler Busy %
Capacity Scheduler	[memory-mb (unit=M), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>	0	0

Show 20 v entries

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU VCores	Allocated Memory MB	Allocated GPUs	Reserve CPU VCores
No data available in table																
Showing 0 to 0 of 0 entries																



Nodes of the cluster

Logged in as: drago

- Cluster
- About
- Nodes
- Node Labels
- Applications
- NEW
- NEW SAVING
- SUBMITTED
- ACCEPTED
- RUNNING
- FINISHED
- FAILED
- KILLED
- Scheduler
- Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used Resources	Total Resources	Reserved Resources	Physical Mem Used %	Physical VCores Used %
0	0	0	0	0	<memory:0 B, vCores:0>	<memory:16 GB, vCores:16>	<memory:0 B, vCores:0>	7	0

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes	Shutdown Nodes
2	0	0	0	0	0	0

Scheduler Metrics

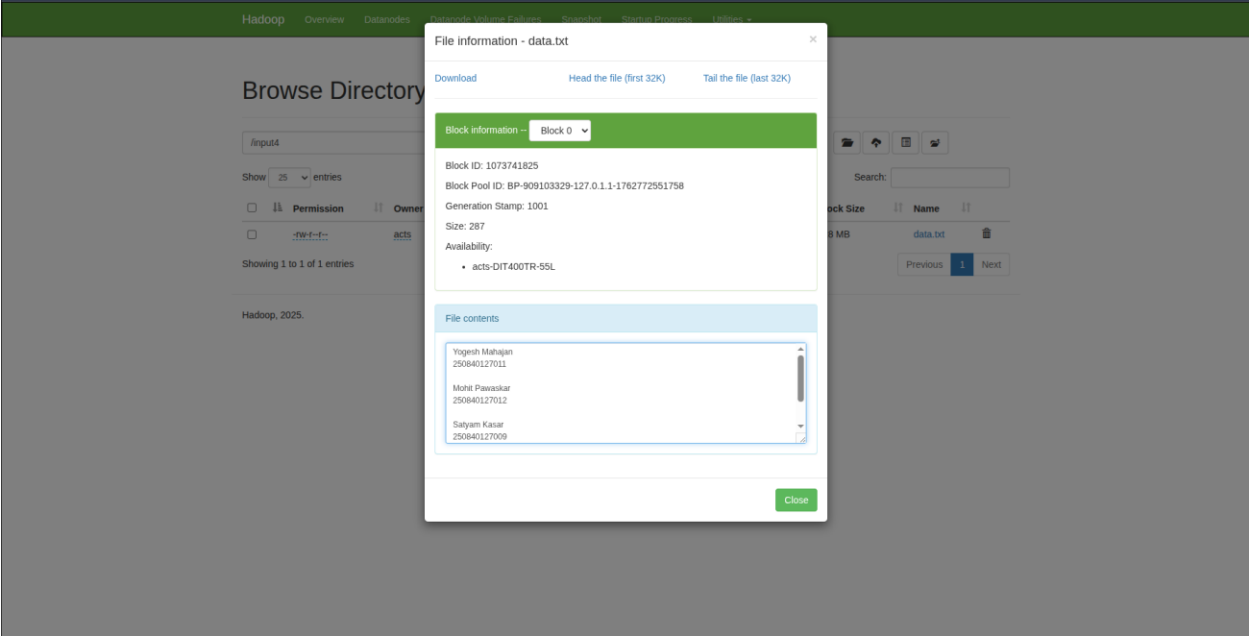
Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation	Maximum Cluster Application Priority	Scheduler Busy %	RM Dispatcher EventQueue Size	Scheduler Dispatcher EventQueue Size
Capacity Scheduler	[memory-mb (unit=M), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>	0	0	0	0

Show 20 v entries

Node Labels	Rack	Node State	Node Address	Node HTTP Address	Last health-update	Health-report	Containers	Allocation Tags	Mem Used	Mem Avail	Phys Mem Used %	VCores Used	VCores Avail	Phys VCores Used %	Version
/default-rack		RUNNING	acts-DIT400TR-55L-37771	acts-DIT400TR-55L-8042	Mon Nov 10 23:02:53 +0530 2025		0		0 B	8 GB	6	0	8	1	3.4.2
/default-rack		RUNNING	acts-DIT400TR-55L-34785	acts-DIT400TR-55L-8042	Mon Nov 10 16:32:33 +0530 2025		0		0 B	8 GB	7	0	8	1	3.4.2

Showing 1 to 2 of 2 entries

First Previous 1 Next Last



```
2025-11-10 16:43:13.407 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
cat: '/input4': Is a directory
cat: '/data.txt': No such file or directory
acts@acts-TP12XH-L2I:/$ hdfs dfs -cat /input4/data.txt
2025-11-10 16:43:19.499 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Yogesh Mahajan
250840127011

Mohit Pawaskar
250840127012

Satyam Kasar
250840127009
~
~
acts@acts-TP12XH-L2I:/$
```